

HYDROFARMS AGRI SOLUTIONS

UNDERGROUND TANK ROOMS, WATERPROOFING AND DRAINAGE

Activity-Specific Construction Method Statement

Document field	Controlled value
Document code	HF-GH01-MST-CIV-003
Revision	00 - Draft for coordination
Project	One-Feddán NFT Lettuce Greenhouse Project
Location	Alexandria, Egypt
Prepared by	Hydrofarms Agri Solutions
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Parent methods	HF-GH01-MST-GEN-002 and HF-GH01-MST-CIV-001

CONTROLLED CIVIL PACKAGE
This method covers the buried chambers and their permanent waterproof envelope. Plastic tanks, pumps, chillers, process piping, cables and controls remain in their respective supply packages; this method controls the civil interfaces required to receive them.

Document Control

Approval and responsibility

Role	Primary responsibility
Project manager	Resources, programme, subcontractor coordination and approval status.
Construction manager	Work-front sequence, temporary works and daily release.
Civil/structural engineer	AFC details, reinforcement, masonry, concrete and retaining action.
Geotechnical engineer	Formation, groundwater, lateral soil condition and backfill recommendations.
Surveyor	Room coordinates, excavation/base levels, openings and as-built records.
QA/QC engineer	ITP, material approvals, concrete/masonry/waterproofing records and tests.
HSE manager	Excavation safety, access, lifting, confined-space controls and edge protection.
MEP engineer	Sleeves, puddle flanges, drains, pump bases and equipment clearances.
Client/consultant	Hold/witness inspection and technical acceptance.

Revision history

Rev.	Date	Purpose	Description
00	15 Aug 2026	Draft for coordination	Initial project-specific English issue

1. Purpose

This method defines the controlled construction, waterproofing, protection, drainage and handover of open-top underground rooms housing the project water tanks. It prevents soil collapse, groundwater ingress, contaminated drainage, tank damage and inaccessible equipment interfaces.

2. Scope and Package Boundary

The civil work package comprises the following controlled activities and permanent interfaces:

- Survey release, excavation interface, formation preparation, blinding and reinforced-concrete bases.
- Reinforced solid concrete-block walls, reinforced-concrete columns/tie beams, internal divider and top edge beams.
- Construction joints, waterstops, sleeves, puddle flanges, waterproofing, protection boards and controlled backfill.
- Sumps, gravity drainage provisions, ladders, guardrails, lifting-beam supports and permanent safe access.
- Civil bases, housekeeping pads and embedded items needed by tanks, pumps and immersed SS316L coils.
- Inspection, water-tightness verification, as-built survey and civil handover to MEP installation teams.

NO DUPLICATION

Tank supply, pump skids, five 7.5 TR chillers, process valves/pipes, sump pumps, power cabling and controls are excluded from this civil package except for their sleeves, supports, bases and interfaces.

3. Governing Information

Priority	Controlled information
1	Approved geotechnical report, groundwater assessment and formation recommendations
2	AFC structural drawings, reinforcement schedules and retaining-wall calculations
3	AFC coordinated tank-room plans, sections, openings, drainage and external works
4	Approved tank, pump, chiller, coil and pipe supplier data
5	HF-GH01-MST-CIV-001 excavation/backfill and HF-GH01-MST-GEN-002 survey releases
6	Approved material submittals, waterproofing method and ITP

STRUCTURAL CONTROL

Room dimensions below are project design inputs. Reinforcement, tie-element spacing, wall reinforcement and bearing details shall be exactly as shown on the approved structural drawings; this method does not replace structural design.

4. Approved Room Schedule

Room / system	Clear internal size (m)	Qty	Base	Wall system	Compartments
Production operating tanks	7.80 x 4.05 x 3.10	1	200 mm RC	250 mm reinforced solid block	2
Reserve-water tanks	12.80 x 3.70 x 3.00	1	200 mm RC	250 mm reinforced solid block	1
Nursery operating tank	3.10 x 3.35 x 2.10	1	150 mm RC	200 mm reinforced solid block	1
Production cooling-pad tanks	4.00 x 4.00 x 3.50	4	200 mm RC	250 mm reinforced solid block	1 each
Service greenhouse pad tank	2.50 x 2.50 x 2.50	1	150 mm RC	200 mm reinforced solid block	1

The eight physical rooms provide nine operating compartments because the production room is divided internally for the independent eastern and western 10 m³ nutrient tanks. The reserve room contains four adjacent 10 m³ tanks in one row. The nursery room receives the confirmed 3 m³ tank. Production cooling-pad rooms receive one 10 m³ plastic tank each; the service pad room receives one 1 m³ tank.

Southern alignment and interfaces

- Moving along the southern fan side: two production operating-tank compartments are centered on the production greenhouse, followed by the reserve-water room, then the nursery room; the WTP is farther west outside the service greenhouse near its fan side.
- Chillers are installed behind the fan side under small independent sunshades; their foundations and maintenance paths shall not obstruct guardrails, tank removal or pipe corridors.
- The reserve room receives treated water from the WTP and transfers it to the two production tanks, the nursery tank and the internal germination tank through the coordinated water-system package.

5. Materials and Permanent Works

Component	Minimum technical requirement	Acceptance evidence
Blinding	Lean concrete, minimum 75 mm or AFC detail, level and clean	Delivery ticket and level survey
Structural concrete	Approved durable reinforced-concrete mix suitable for below-grade exposure	Mix approval, cubes and pour record
Reinforcement	High-yield deformed steel to approved bar schedule; clean and correctly supported	Mill certificates and pre-pour inspection
Buried wall units	Dense solid concrete masonry units; 250 mm or 200 mm as scheduled	Source approval and strength certificates
Grout/mortar	Approved compatible mix; reinforced cells fully grouted where detailed	Batch/control records
Waterstop	PVC/hydrophilic type compatible with joint and groundwater condition	Product approval and joint inspection
Waterproofing	Approved below-grade membrane/system compatible with substrate and expected water pressure	Manufacturer submittal and applicator approval
Protection	Protection board or screed preventing puncture during backfill	Concealed-work inspection
Sleeves/seals	Rigid sleeves with puddle flange or proprietary watertight link seal	Opening schedule and pressure/water test
Metalwork	Hot-dip galvanized ladder, guardrail, toe-board and lifting supports where shown	Coating certificate and inspection

PROHIBITED WALL MATERIAL

Lightweight AAC/white block shall not be used as a buried retaining wall. It may only be used for non-loadbearing above-ground partitions. Buried rooms use reinforced dense solid concrete masonry tied into reinforced-concrete elements, or a fully reinforced-concrete alternative approved by the structural engineer.

6. Preconditions

- Approved excavation, structural, opening and MEP coordination drawings are available at the work front.
- Formation and excavation stability have been accepted under HF-GH01-MST-CIV-001.
- Tank vendor data confirm actual diameter/height, lifting method, nozzle coordinates and minimum maintenance clearances.
- All sleeves are taken from one coordinated opening schedule; core drilling through completed waterproofed walls is prohibited without written approval.
- Waterproofing substrate requirements, corner details, joint treatment and termination levels are approved.
- Temporary dewatering, safe access, edge protection, lifting route and emergency arrangements are operational.

7. Detailed Execution Method

7.1 Survey and formation release

- Verify room centerlines, internal clear dimensions, wall faces, base top level, pipe inverts and adjacent equipment coordinates.
- Inspect formation with the civil/geotechnical team. Remove disturbed or unsuitable soil and reinstate with approved compacted fill or lean concrete.
- Keep the formation dry and place blinding promptly after acceptance.

7.2 Blinding, kicker and base reinforcement

- Place and finish blinding to provide a clean, stable waterproofing/reinforcement working surface.
- Set out wall, divider, sump, sleeve and embedded-item lines on the blinding; check diagonals and offsets.
- Install the specified under-base waterproofing where the selected system requires a continuous tanking envelope; protect it before reinforcement work.
- Fix reinforcement, chairs, starter bars, waterstops, sump reinforcement, sleeves and embedded plates to the AFC schedules.
- Use non-corroding spacers of approved cover. Welding reinforcement or puncturing the membrane is prohibited unless detailed and approved.

7.3 Base slab concreting

- Complete the pre-pour hold-point inspection covering dimensions, cover, cleanliness, joints, waterstops, sleeves and access.
- Pour continuously by an approved sequence, compact without displacing reinforcement or waterstops and finish to the tank-support tolerance.
- Form the sump and drainage falls without ponding. Cure concrete by the approved method and protect edges/starters.
- Do not load the slab or begin wall construction until the specified concrete strength and release are achieved.

7.4 Reinforced masonry walls and RC ties

- Set dense solid units to line and level on the released base. Keep cavities, starters and reinforcement clean.
- Install vertical bars, RC columns, horizontal bond/tie beams and top edge beam exactly to the structural drawing.
- Fully grout reinforced cells/lifts by the approved sequence; prevent voids and clean grout from sleeve faces and waterstops.
- Construct the production-room divider as a complete independent hydraulic/operational separation while preserving the approved access and pipe interfaces.
- Provide formed recesses and cast-in supports for ladders, guardrails, pipe brackets and the approved 1-ton manual chain-block beam where scheduled.

7.5 Openings, sleeves and penetrations

Interface	Civil provision	Release condition
3-inch feed mains	Coordinated sleeve/puddle flange with installation tolerance	Pipe route and tank nozzle verified
6-inch gravity returns	Sleeve at controlled invert; no reverse fall or reduced clear bore	Invert survey and dry fit
Reserve transfer lines	Sleeves for WTP inlet and operating/nursery tank transfer branches	Valve access and route verified
Cooling-pad circuits	Pad-water feed/return sleeves separated from structural reinforcement	Sector routing verified
Electrical/controls	Dedicated sleeves above drainage risk zone where practicable	Electrical schedule released
Drain/sump discharge	Sleeve and sealed penetration to approved discharge route	Drainage route accepted

7.6 Waterproofing substrate preparation

- Allow concrete/masonry to cure as required by the selected system. Remove fins, laitance, dust, oil, loose mortar and sharp projections.
- Repair honeycombing, cracks, tie holes and defective joints with approved repair materials. Form compatible fillets at internal corners.
- Check moisture condition and surface profile against manufacturer limits before primer or membrane application.

7.7 Waterproofing application

- Apply primer, membrane/coating, reinforcement strips and successive layers strictly to the approved manufacturer method and coverage rate.
- Continue the external membrane from base to above final ground termination and integrate it with the under-base system where specified.
- Provide additional detailing at corners, joints, sleeves, puddle flanges, sumps, RC/masonry interfaces and changes in section.
- Internally apply the approved hygienic water-resistant protective coating to room surfaces where scheduled; it is not a substitute for external tanking.
- Record batch numbers, ambient/substrate conditions, coverage, wet/dry film thickness where applicable and applicator names.

7.8 Protection, drainage and backfill

- Inspect and accept the completed membrane before covering. Repair test areas and defects using the approved compatible detail.
- Install protection boards/drainage layer without puncturing the membrane. Keep termination bars and seals accessible for inspection.
- Provide one formed sump in each room/compartment as scheduled, with floor falls toward it and a removable corrosion-resistant cover.
- Backfill symmetrically in loose lifts not exceeding 200 mm using approved selected material and small equipment near walls.
- Achieve at least 95% laboratory maximum dry density around structural rooms unless the geotechnical design requires a higher value.
- Do not create unbalanced earth pressure, impact walls, bury untested penetrations or cover membrane damage.

7.9 Permanent access and edge safety

- Install hot-dip galvanized ladders with secure handholds, non-slip rungs and clear landing space.
- Provide continuous guardrails, midrails and toe boards around open tops except at controlled removable access sections.
- Use lockable removable barriers/covers where maintenance openings interrupt the guardrail.
- Keep tank lifting/removal zones and the manual chain-block path clear of permanent pipework and sunshade supports.

8. Inspection and Test Plan

No.	Inspection stage	Method	Type	Responsibility	Record
01	AFC drawings, survey and opening schedule	Document review	H	Civil/MEP/QA-QC	Work-front release
02	Formation and blinding level	Survey/visual	H	Civil/Geotech/Consultant	Formation release
03	Under-base waterproofing, if specified	Visual/system test	H	QA-QC/Applicator	Membrane release
04	Base reinforcement, joints and sleeves	Pre-pour inspection	H	Civil/MEP/Consultant	Pour card
05	Concrete placement and strength	Site/lab testing	W	Civil/QA-QC/Lab	Concrete report
06	Masonry, reinforcement and grouting	Visual/material tests	W	Civil/QA-QC	Wall inspection
07	RC ties/top beam and embedded items	Pre-pour inspection	H	Civil/MEP/Consultant	Pour card
08	Sleeves and penetration seals	Survey/visual/test	H	MEP/QA-QC/Consultant	Opening release
09	Waterproofing substrate and details	Visual/moisture check	H	Applicator/QA-QC	Substrate release
10	Completed waterproofing	Coverage/DFT/holiday test	H	QA-QC/Consultant	Membrane report
11	Protection and drainage before backfill	Visual/function test	H	Civil/MEP/Consultant	Backfill release
12	Backfill lifts and density	Field density test	W	QA-QC/Lab	Compaction report
13	Guardrails, ladders and lifting supports	Dimensional/load check	W	Civil/HSE/Consultant	Metalwork release
14	Final as-built and civil handover	Survey/record review	H	QA-QC/Client	Turnover dossier

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

9. Hold Points

- HP-TR-01: Formation, coordinates and clear excavation footprint accepted before blinding.
- HP-TR-02: Base reinforcement, waterstops, sleeves, sump and embedded items accepted before concrete.
- HP-TR-03: Wall reinforcement, tie elements and openings accepted before grouting/concrete.
- HP-TR-04: Waterproofing substrate and penetration details accepted before membrane application.
- HP-TR-05: Completed waterproof envelope tested and accepted before protection/backfill.
- HP-TR-06: Buried pipework tested and as-built surveyed before penetrations/routes are concealed.
- HP-TR-07: Civil room, safety metalwork, drainage and records accepted before tank/equipment installation.

10. Acceptance Criteria

Characteristic	Acceptance requirement
Internal dimensions	Not less than scheduled clear dimensions; tank and maintenance clearances maintained
Base level/flatness	Within AFC tolerance and tank manufacturer support requirement; no rocking or point support
Wall alignment	Plumb, true and within structural/masonry tolerance; no harmful bowing or open joints
Concrete	Specified strength achieved; defects assessed and repaired by approved method
Waterproofing	Continuous, bonded and defect-free; penetrations/joints complete; required tests passed
Drainage	Positive fall to sump, no uncontrolled ponding, discharge route functional
Backfill	Approved material, symmetrical placement, maximum 200 mm loose lifts and minimum 95% MDD
Access/safety	Secure ladder, complete guardrail/toe board and unobstructed lifting/maintenance route
Interfaces	All sleeves/nozzles/bases tagged, dimensionally checked and matched to equipment/shop drawings

11. HSE and Environmental Controls

Hazard	Mandatory control	Residual risk
Excavation collapse	Approved slope/shoring, competent-person inspections and controlled access	Medium
Fall into open room	Rigid barriers, guardrails, lighting and signed access	Medium
Confined-space entry	Assessment, permit, gas test, ventilation, standby and rescue plan when criteria apply	Medium
Lifting tanks/materials	Lift plan, certified gear, exclusion zone and competent operators	Medium
Waterproofing chemicals	SDS controls, ventilation, PPE, ignition control and approved storage	Low
Dewatering/electrical	Protected supply, earthing/RCD, dry connections and standby pumping	Medium
Hot work/galvanized steel	Permit, fire watch, fume control and coating repair	Low
Contaminated discharge	Sediment control and discharge only to the approved route	Low

12. Records and Handover

- Approved material submittals, mix designs, reinforcement/masonry certificates and waterproofing warranties.
- Survey releases for formation, base, walls, openings, pipe inverts and final as-built coordinates.
- Concrete delivery/cube records, masonry/grout inspections and repair approvals.
- Waterproofing substrate, batch, coverage, DFT/holiday/flood-test records and concealed-work photographs.
- Backfill source approvals, lift register and field-density test reports.
- Metalwork coating certificates, access inspection and lifting-support verification.
- Signed ITP, NCR closeout and room-by-room civil handover certificate to the MEP installation team.

Appendix A - Room Release Checklist

- Room code, coordinates, clear dimensions and top/base levels match the approved drawings.
- Tank route and lifting/removal envelope remain unobstructed.
- All sleeves are tagged; 3-inch feeds and 6-inch returns are at the approved location and invert.
- Waterproofing and penetration seals are accepted and protected.
- Sump, falls and discharge operate without uncontrolled ponding.
- Ladder, guardrail, toe board and removable access barrier are complete.
- No sharp projections, debris or unsupported pipework can damage the plastic tank.
- As-built survey, photographs and test dossier are attached to the release.

NEXT METHOD STATEMENT

The next coordinated submission is HF-GH01-MST-CIV-004 for buried utilities, sleeves, pipe bedding, testing and route reinstatement. Its approved pipe profiles and opening schedule shall be coordinated with this method before any wall or base is cast.